

Optimum unit sizing of hybrid renewable energy system utilizing harmony search, Jaya and particle swarm optimization algorithms

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ABSTRACT

Renewable energy technologies, which include wind turbines, biomass and photovoltaics, are extensively utilized in the generation of electricity in an island zones. Stand-alone systems that incorporate hybrid sources of renewable energy are employed and require different sizing of components to obtain optimum units to reduce fundamental challenges. In this paper, the Harmony Search (HS), Jaya and particle swarm optimization (PSO) algorithms are utilized in sizing an optimum design of a hybrid renewable energy system (HRES) containing a set of the Wind-Photovoltaic-Biomass-Battery technologies with a goal of cost-effectively, efficiently, and reliably meeting customers' electricity demands.

The outcomes of the three algorithms are compared, and the one with the most techno-economic optimum unit sizing of the HRES was determined. The hybrid system's reliability and efficiency are calculated using two factors: the maximum allowable loss of power supply probability ($LPSP_{max}$) together with reduced allowable excess energy fraction (EFF_{min}). Different configurations consisting of solar PV, biomass generators, wind turbines, and batteries are modeled, simulated and optimized to obtain a system combination that is most energy efficient and cost effective in the studied zone. The study demonstrated that the ideal system with the least cost and the best performance was that which consists of thirteen solar PV systems (70.98 kW), four biomass systems (160 kW), one wind turbine (20 kW) and 15 Ni-Fe battery banks (288 kW h), with a total system present cost of \$581,218 and a 0.254 \$/kWh cost of energy.

1. Introduction

The steady increase in the global population in recent years has given rise to higher energy consumption for both industrial and residential applications. As a result, there is a need to generate more electricity to curb skyrocketing demand. According to Bloomberg New Energy Finance, the global electricity demand is expected to reach approximately 38,700 TW-hours (TWh) by 2050, which is up from 25,000 TWh in 2017, and the anticipated increase is driving significant new investment in power generating capacity (Hill, 2018). Currently, over 60 % of global electricity is produced using fossil fuels and their continued utilization yields greenhouse gases, some of which are responsible for degradation of the ozone layer that has led to global warming and is detrimental to the environment (Das, Singh, & Biswas, 2019; Konneh et al., 2019; Sedaghati et al., 2018). In order to cater to the increasing electricity demand as well as to minimize environment degradation, distributed energy systems (DES), which include solar, wind, biomass, petrochemical, and fuel cell among others, have been widely used (Gharibi & Askarzadeh, 2019). Numerous researchers have

shown that the uptake of these technologies have been on the rise especially in the non-urban regions where grid electricity is inaccessible.

The dominant sources of energy used in DES are renewable energy technologies, particularly wind, solar, and biomass. Despite the desire to accelerate the use of renewable energy sources, most types depend on weather conditions which minimize their individual efficiency and reliability while increase system costs. To improve the system, a multi-objective approach must be developed that aims at fostering reliability, cost effectiveness, and efficiency along with developing a renewable energy index. Thus, different technologies should be combined to form a hybrid renewable energy system (HRES) where the elements complement each other (Samy, Barakat, & Ramadan, 2019; Samy, Barakat, & Ramadan, 2019; Samy, Sarhan, Barakat, & Al-Ghamdi, 2018). Even so, the HRES has to be designed to operate optimally to work in harmony with other technologies in order to realize its full potential. As such, the designing or sizing of an HRES that is technically sound, cost-effective, and operates optimally is a crucial phase in the development of such a system.

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Acronyms

P_s	PV output power in (kW)	V_{ci}	The cut in wind speed in (m/s)
P_{PVR}	Rated module output in (kW)	V_{co}	The terminate speed of wind (m/s)
N_{pv}	Number of PV modules	V_r	Estimated wind speed in (m/s)
PV_{df}	PV derating factor (0.8)	H_{WT}	Wind turbine hub altitude
$G(t)$	The overall irradiation circumstance on the tilted plane (kW/m ²)	E_{Bat}	Battery bank nominal capacity (kWh)
G_{base}	The Solar irradiation at standard status in (1 kW/m ²)	$E_{Bat,min}$	Minimum allowable storage battery capacity
K_T	Temperature coefficient of maximum power ($K_T = -3.7 \times 10^{-3}(1/^\circ C)$) for Polly and Monocrystalline silicon	$E_{Bat,max}$	Maximum allowable storage battery capacity
$T_{amb}(t)$	Ambient air temperature in ($^\circ C$)	LPSP	Loss of Power Supply Probability
T_{base}	The temperature at reference condition ($^\circ C$)	C_{NPC}	The net present value
NOCT	The normal operating cell temperature ($^\circ C$)	N_{bat}	The number of battery banks
P_{bio}	The biomass gasifier output power (kw)	EEF	Excess energy Fraction
Feedstock	The hourly rate of the biomass feedstock in (kg/h)	N_{proj}	Project lifetime (years)
HHV	The higher heating values of the biomass feedstock (kJ/kg)	C_I	The initial capital cost
η_g	The gasifier gasification efficiency (75 %)	C_R	The replacement costs of the components
E_{bio}	The yearly energy produced from the biomass system	$C_{O\&M}$	Operating and maintenance costs
N_{gen}	The number of biomass systems	S	Salvage value
P_{WT}	Wind turbine output power	C_F	The Fuel cost
N_{WT}	The number of wind turbines	I_r	Real interest rate
η_w	The wind turbine efficiency	N_{comp}	Component lifetime (years)
P_{WTr}	The wind turbine estimated power (kW)	INT	The function applied to return the integer amount of a real number
$V(t)$	The evaluated wind speed in (m/s)	C_{bio}	The biomass fuel cost
		M_{total}	Overall wood consumption biomass generator (kg/y)
		COE	Cost of generating energy (\$/kWh)
		E_{served}	Primary load served (kWh/year)

The indiscriminate availability of renewable sources makes the sizing of an HRES very complicated and includes a trade-off among cost-effectiveness and reliability aspects. Thus, numerous models (algorithms) and software tools have been employed in optimizing HRES designs and more tools continue to be developed. Some of the utilized software tools include RETScreen, HybSim and the Hybrid Optimization of Multiple Energy Resources (HOMER) (Sinha & Chandel, 2014). The last tool is commonly and extensively used to validate optimized results that are mostly based on the cost of energy and the technical integration of system equipment like wind turbines, solar modules, inverters and controllers, biogas generators, and energy storage systems. In (Elkadeem et al., 2020) and (Elkadeem, Wang, Sharshir, & Atia, 2019), the authors combined technical, environmental and economic aspects in HOMER Pro software to formulate a decision-making framework that efficiently plans and assesses hybrid renewable energy based microgrid (HRE-MG) systems. The net present value (NPV) was utilized as the most reliable metric when evaluating the best combination of different subsystems based on their technical, environmental and economic performances to form the HRE-MG. Then, the systems' configurations were ranked based on the total life cycle cost (also known as NPV) and the least one was selected as the optimum. Also, in (Odou, Bhandari, & Adamou, 2020) and (Islam, 2018), HOMER software was used to rank technically feasible and environmentally friendly distributed energy system configurations based on overall annualized costs as well as levelized cost of energy (LCOE).

While software tools are readily available and easy to use in optimizing HRES (Sinha & Chandel, 2014), the design and sizing procedures involved do not permit customization of the design based on user defined constraints. Thus, to mitigate this drawback, mathematical models and algorithms are usually applied in the modern HRES design process.

The utilization of Meta-heuristic algorithms in design and sizing is the modern technique of optimizing an HRES. There are numerous algorithms, but the renowned meta-heuristic ones include particle swarm optimization (PSO), evolutionary metaheuristic, simulated annealing, differential evolution, genetic algorithm (GA), cuckoo search, and the

hybrid PSO simulation-based metaheuristic algorithm among numerous others. In (Kumar et al., 2017), the five important criteria of technical, social, economic, political/institutional, and environmental were used to measure sustainability of a microgrid to aid in decision making. Similarly, an Analytic Hierarchy Process (AHP) was used in (Nsafon et al., 2020) to determine the importance of the criteria to be employed to obtain the optimized hybrid power systems. In (Abdmouleh, Gastli, Ben-Brahim, Haouari, & Al-Emadi, 2017), a multi criteria decision analysis (MCDA) was used to integrate renewable energy into the utility grid by analyzing techno-economic optimization for seasonal changes to obtain the best configuration using the AHP framework.

In general, the methods broadly applied to obtain the unit sizing solution for HRES are the particle swarm optimization (PSO) technique and HOMER software package. In (Singh, Singh, & Kaushik, 2016), in which the authors proposed an algorithm known as artificial bee colony (ABC) to conduct a viability study for the HRES that consisted of wind, biomass and photovoltaic (PV), then compared the results of ABC to those of both HOMER and PSO. Study in (Paliwal, Patidar, & Nema, 2014) presents a systematic approach to determine the suitable size of an autonomous system of hybrid power. The research used the PSO in finding the optimum sizing of a component. Previous, a multi-faceted PSO technique used to determine optimal sizes of hybrid systems relied on the ϵ -constraint technique, which minimizes system aspects such as the unmet load estimate as well as GHG emissions (Sharafi & ElMekkawy, 2014). In addition, the study considered an economic assessment of an island PV/Biomass hybrid system that considered different types of battery-manufactured technologies such as the lithium ferro phosphate (LFP) flooded with lead-acid (FLA) and nickel iron (Ni-Fe) (Eteiba, Barakat, Samy, & Wahba, 2018). The outcomes showed the nickel iron battery to be the best choice for off-grid applications.

In the case of unit sizing, the genetic algorithm (GA) has been used because of its merits, which include transacting with different objectives, flexibility to conform, acceptable convergence, and efficiency in comparisons to the authorized tactics (Ismail, Moghavvemi, & Mahlia, 2014). In addition, the genetic algorithm (GA) efficiently and

dynamically addressed non-linear problems because of its abilities to search the scope to obtain an optimum result. The genetic algorithm was utilized to mitigate energy estimates in (Chen, Stephens, Koon, Ashtine, & Koon, 2020; Sharief & Sairam, 2019) and reduced the annual charge of off-grid systems that contained PV/wind/micro-turbine hybrid systems with battery storage.

Previously, various studies employed different meta-heuristic algorithms in HRES sizing including the Artificial Bee Colony (ABC) (Govardhan & Roy, 2015; Kefayat, Lashkar Ara, & Nabavi Niaki, 2015; Sundareswaran, Sankar, Nayak, Simon, & Palani, 2015), Flower Pollination (FP) (Barakat, Samy, Eteiba, & Wahba, 2016; Barakat, Samy, Eteiba, & Wahba, 2017; Dubey, Pandit, & Panigrahi, 2015; Sudabattula & Kowsalya, 2016), and the Harmony Search (HS) (Cheng, Bae, & Horton, 2019; Golmohamadi, Keypour, Bak-Jensen, & Pillai, 2019; Mosaad, El-Raouf, Al-Ahmar, & Bendary, 2019; Najafi-Ghalelou, Zare, & Nojavan, 2019) among others. The aforementioned techniques, as well as other meta-heuristic ones, required specific conditions such as common control, particular algorithmic variables, and tuning in order to produce optimum HRES solutions.

One of the objectives of the presented research is to make a comparison between three types of optimization techniques, which are the HS, Jaya, and PSO algorithms. These algorithms were chosen because they are examples of different optimization techniques. PSO and HS are examples of metaheuristic optimization techniques while the Jaya algorithm, which is not based on any natural phenomena, can be considered as a non-algorithmic specific parameter which always tries to get closer to success and avoid failure. The Harmony Search is distinguished by the fact that it does not need preparation for the initial values, and it is possible to use discrete and continuous variables that

may escape from the local optima. In fact, PSO was frequently used in the literature because PSO has the advantage of having higher probability and efficiency in finding the global optima but has a disadvantage of converging prematurely and becoming trapped in a local minimum, especially with complex problems.

In this paper, HS, Jaya, and PSO algorithms are used in sizing an optimum design of a HRES containing a combination of Wind-PV-Biomass-Battery technologies with a goal of determining cost-effectiveness, efficiency, and reliability to better meet customers' electricity demands. The HRES reliability and efficiency are calculated using two factors: the maximum allowable loss of power supply probability ($LPSP_{max}$) and the least allowable excess energy fraction (EFF_{min}). The performance of HS and Jaya optimization algorithms is estimated and compared to the PSO in order to determine the optimal technological and economical design of the hybrid system. Recently, the Particle Swarm Optimization (PSO) is regarded as a benchmark algorithm.

This study addresses the following objectives:

- Illustrate in specifics the modeling of the PV/wind/ biomass HRES system.
- Develop a model for the calculation of the electric energy generated by a biomass generator based on the estimation of the higher heating value (HHV) of biomass feedstock.
- Present a power management strategy that regulates the power flow among such system equipment.
- Perform various sensitivity analyses on an optimum system to forecast the system's next execution.

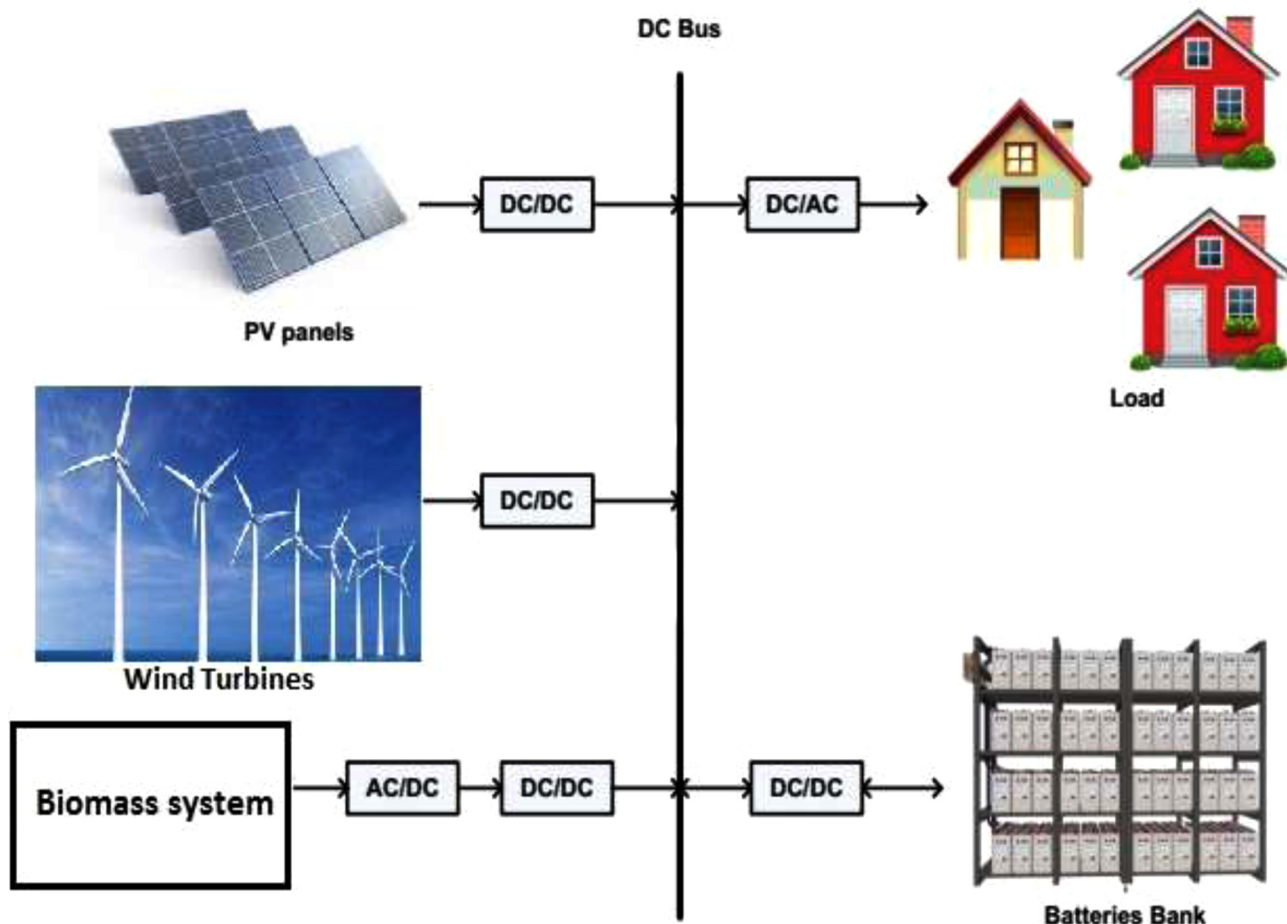


Fig. 1. Proposed system model for the stand-alone HRES.

2. Modeling of HRES elements

In this section, equations associated with each element are presented. Definition of the problem's objective function and energy management strategy is revealed. Fig. 1 indicates the proposed system constituted from solar photovoltaic, Biomass Gasifier with syngas machine, wind turbine, batteries, dump load, and an inverter.

2.1. PV generator modelling

For the correct simulation of the PV power generator performance, it is necessary to provide readings of solar radiation falling on the study area for one year in steps of one hour. The daily information of the solar irradiation and the utilized temperature for the study zone were collected from the HelioClim-3 database (SoDa Website, 2019).

Expression of the PV generator output power is as shown below:

$$P_s = \sum_{t=1}^{8760} (P_{PVr} N_{pv} P_{Vdf}) \left(\frac{G(t)}{G_{base}} \right) \left(1 + K_T \left(T_{amb}(t) + G(t) \times \left(\frac{NOCT - 20}{800} \right) * 1000 \right) - T_{base} \right) \quad (1)$$

2.2. Biomass generator modelling

The analysis in this paper focused on utilizing crop residues from a farm. The biomass generator is used as a backup generator to be operated if the PV's power output and wind turbine generators are not sufficient to meet electrical load demand at a particular time.

The amount of power produced by the gasifier at any hour can be calculated as:

$$P_{bio}(t) = Feedstock(t) \times HHV \times 27.78 \times 10^{-5} \times \eta_g \quad (2)$$

Where the number (27.78×10^{-5}) is a factor for converting units from kJ to kWh dimensionless. The proposed model computes the HHV according to the biomass' chemical composition (Friedl, Padouvas, Rotter, & Varmuza, 2005).

$$HHV = 3.55C^2 - 232C - 2230H + 51.2C \times H + 131N + 20600 \quad (3)$$

The generator operates at a 30 % load ratio from its rated capacity as this will prevent it from running at considerably lower demands. The ultimate load ratio of 80 % is to be considered from its evaluated capability.

Let

G_{min} = 30% of the generator rated capacity

G_{max} = 80% of the generator rated capacity

G_{output} = the output power of the generator

Then

$$G_{output} = \begin{cases} 0 & P_{bio} < G_{min} \\ G_{max} & P_{bio} > G_{max} \\ P_{bio} & G_{max} < P_{bio} < G_{min} \end{cases} \quad (4)$$

The yearly energy produced E_{bio} can be evaluated as:

$$E_{bio} = \sum_{t=0}^{8760} N_{gen} G_{output} \times t \quad (5)$$

2.3. Wind turbine modelling

The output power of wind turbine generators mainly relies on the wind speed. NASA provides the data on wind speed every month that has been used as input data (NASA, 2020).

The estimate of the output power of the wind turbine is as indicated below:

$$P_{WT} = \begin{cases} N_{WT} \times \eta_w \times P_{WTr} \times \sum_{t=1}^{8760} \left(\frac{V(t)^3 - V_{ci}^3}{V_r^3 - V_{ci}^3} \right), & V \leq V_r \\ N_{WT} \times \eta_w \times P_{WTr}, & V_r \leq V \leq V_{co} \\ 0, & V_{co} \leq V \text{ or } V \leq V_{ci} \end{cases} \quad (6)$$

Where:

$V(t)$ - Rated wind speed in (m/s) at hub altitude.

$$V(t) = V_r \left(\frac{H_{WT}}{H_r} \right)^\alpha \quad (7)$$

Where:

H_{WT} is the wind turbine hub height,

H_r is the base altitude, and

α is the friction parameter and has a value of (1/7) for surfaces with minimal roughness and wide-open site.

2.4. Modeling of the battery bank system

Batteries store the system's excess capacity of energy in order to be used at the time of need. The battery energy storage is subjected to the following constraints:

$$E_{Bat,min} \leq E_{Bat}(t) \leq E_{Bat,max} \quad (8)$$

$$E_{Bat,max} = E_{Bat,cap} \quad (9)$$

$$E_{Bat,min} = E_{Bat,max}(1 - DOD) \quad (10)$$

2.5. Reliability criteria

In stand-alone HRES, reliability is an important factor, and is represented in this paper as LPSP and EEf.

2.5.1. Loss of power supply probability (LPSP)

The loss of power supply probability is characterized by a number between zero and one, where value one implies that the load will be rejected, and zero value shows that the load will always be satisfied.

$$LPSP = \frac{\sum_{t=1}^T P_{deficit}(t) \cdot \Delta t}{\sum_{t=1}^T P_{demand}(t) \cdot \Delta t}, \quad T = 8760 \text{ and } \Delta t = 1. \quad (11)$$

2.5.2. Excess energy fraction (EEf)

$$EEf = \frac{\sum_{t=1}^T \text{wasted energy}(t) \cdot \Delta t}{\sum_{t=1}^T E_{PV}(t) \cdot \Delta t + \sum_{t=1}^T E_{Bio}(t) \cdot \Delta t}, \quad T = 8760 \text{ and } \Delta t = 1. \quad (12)$$

2.6. GHG emission estimation

In this section of the paper, the method of calculating the amount of GHG emissions resulting from the use of the PV system, the wind turbines, and the biomass generator are estimated. Table 1 presented the equations used to evaluate the GHG emissions in tones from the various devices of the existing system.

Table 1

Estimation CO₂ emission from the several devices of the system.

The Emission Source	GHG Emissions (tones)	Emission Factor (kgCO ₂ /kWh)
PV System	$E_{mpv} = \sum_{t=1}^{8760} P_{PV}(t) \times e_{pv}$	0.045 for mono-Si PV (Peng, Lu, & Yang, 2013)
Wind Turbine	$E_{mWT} = \sum_{t=1}^{8760} P_W(t) \times e_{WT}$	0.011 (Ackermann, Andersson, & Söder, 2001)
Biomass	$bio_{GHG} = \sum_{t=1}^{8760} P_{bio}(t) \times e_{bio}$	0.099 (Weisser, 2007)

The village load connected to the grid is considered as a baseline case in comparing the GHG emission with the use of renewable energy resources.

$$\text{Base Case GHG emissions} = \text{Total load} \times \text{Grid electricity-specific factor} \times \text{GWP} \quad (13)$$

where, the electricity-specific factor for KSA is 0.7956 (kgCO₂/kWh)

(Brander et al., 2011).

GHGs per unit emitted have differing degrees of impact on global warming due to the particular property of the gas. Therefore, to compare the emissions of different GHGs, all emissions are referred to as CO₂ equivalents (CO₂e). This CO₂e is a scale where CO₂ is the reference point and has a global warming potential (GWP) of 1, and every other GHG listed in the Kyoto Protocol has a greater GWP compared to CO₂ (Solomon, 2007).

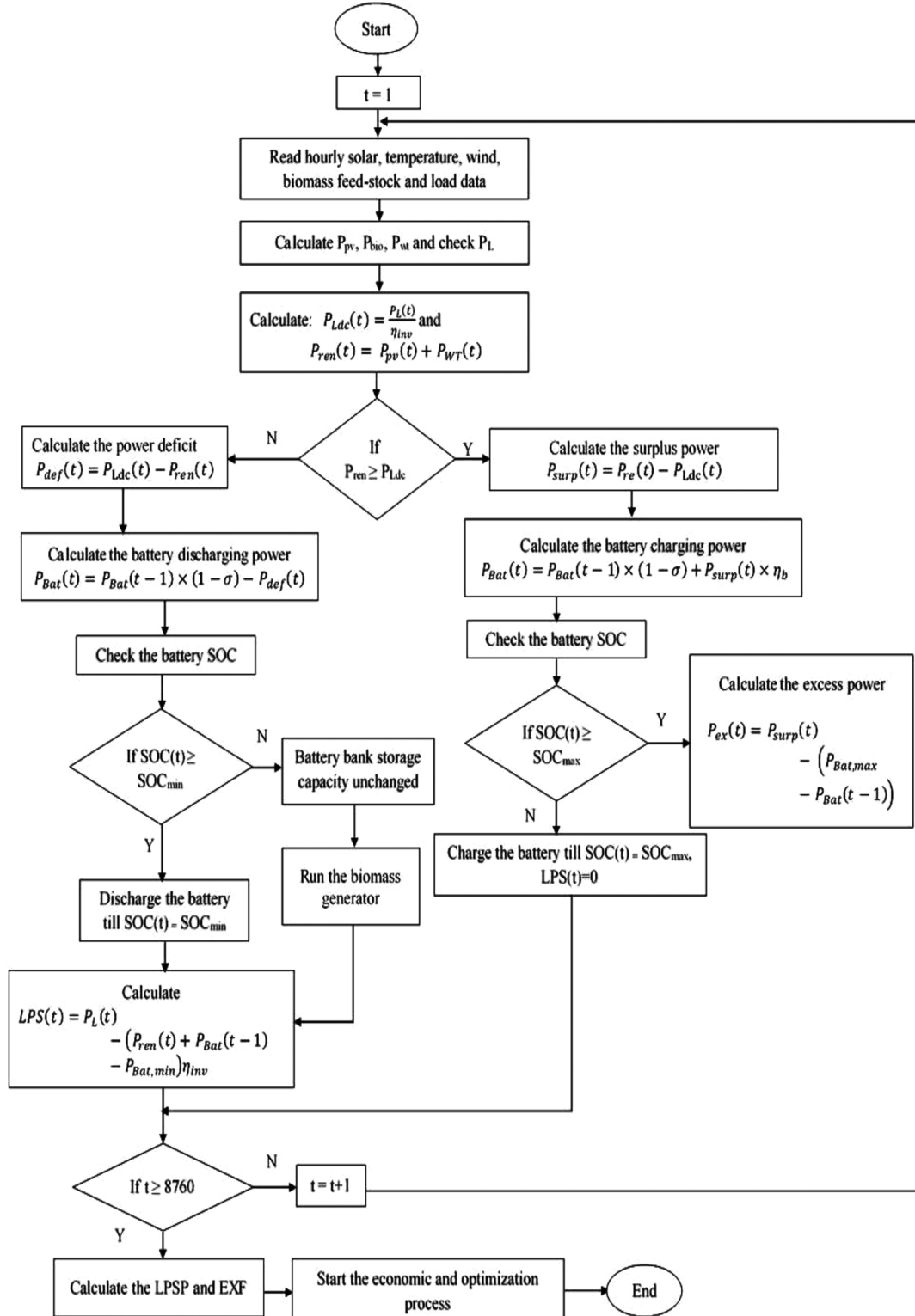


Fig. 2. Flowchart of power arrangement scenario of suggested system.

```

For each particle
  Initialize particle
End For
Do
  For each particle
    Calculate fitness value
    If the fitness value is better than the best fitness value Pbest in history
      Set current value as the new Pbest
  End
  Choose the particle with the best fitness value of all the particles the
  Gbest
  For each particle
    Calculate particle velocity according to the following equation
     $V_{i,d} = \omega \times V_{i,d} + C_1 \times R_1 \times (P_{best,i,d} - X_{i,d}) + C_2 \times R_2 \times (G_{best,i,d} - X_{i,d})$ 
    Update particle position according to the following equation
     $X_{i,d} = X_{i,d} + V_{i,d}$ 
  End
While maximum iterations or minimum error criteria is not attained.

```

Fig. 3. PSO algorithm pseudo-code.

The higher the GWP of a gas, the greater the contribution to the greenhouse effect. Methane has a GWP of 21 and nitrous oxide has a GWP of 310. The default values are defined by International Panel on Climate Change (IPCC) (De Muizon & Glachant, 2004).

The overall CO₂ GHG emission reduction by the system is estimated as:

$$System_{CO_2total}(\text{tons}) = PV_{CO_2GHG} + WT_{CO_2GHG} + bio_{CO_2GHG} \quad (14)$$

$$net_{savingCO_2total}(\text{tons}) = Base\ Case_{CO_2} - System_{CO_2total} \quad (15)$$

3. Objective function and constraints

3.1. Objective function

The aim is to establish an optimal combination of the system components to attain minimum value of the net present cost of the system indicated as (CNPC).

Therefore, CNPC's minimization is expressed using the formula below:

$$\text{Minimize } C_{NPC} \{N_{PV}, N_{bat}, N_{gen}, N_{WT}\} \quad (16)$$

Subject to:

$$N_{PV}^{min} \leq N_{PV} \leq N_{PV}^{max}$$

$$N_{bat}^{min} \leq N_{bat} \leq N_{bat}^{max}$$

$$N_{gen}^{min} \leq N_{gen} \leq N_{gen}^{max}$$

$$N_{WT}^{min} \leq N_{WT} \leq N_{WT}^{max}$$

$$0 \leq LPSP \leq LPSP_{max}$$

$$0 \leq EEF \leq EEF_{max}$$

Computation of the system's net present cost is as follows:

$$C_{NPC} = \sum_x C_I + C_R + C_{O\&M} + C_F - S \quad (17)$$

$$C_{NPC} = \sum_{n=0}^{N_{proj}} (C_I(n) + C_R(n) + C_{O\&M}(n) + C_F(n) - S(n)) \times \frac{1}{(1 + I_r)^n} \quad (18)$$

$$C_I = \sum N_{PV} C_{IPV} + N_{bat} C_{Ibat} + N_{gen} C_{Igen} + N_{WT} C_{IWT} \quad (19)$$

$$C_R = \sum N_{PV} C_{RPV} + N_{bat} C_{Rbat} + N_{gen} C_{Rgen} + N_{WT} C_{RWT} \quad (20)$$

```

Begin
  Define the objective function  $f(X)$ ,  $X = (x_1, x_2, \dots, x_n)$ .
  Define HMCR, PAR, bw and other parameters.
  Generate harmony memory with random harmonies.
  While j < number of iterations
    While l < number of variables
      If (rand < HMCR) choose a value from HM for the variable i
      If (rand < PAR) adjust the value by adding a certain amount
      End If
      else choose a random value
      End If
    End while
    Accept the new harmony if better
  End while
  find the current best solution
End

```

Fig. 4. HS algorithm pseudo code.

$$C_{O\&M} = \sum N_{PV} C_{O\&MPV} + N_{bat} C_{O\&Mbat} + N_{gen} C_{O\&Mgen} + N_{WT} C_{O\&MWT} \quad (21)$$

$$S = \sum N_{PV} S_{PV} + N_{bat} S_{bat} + N_{gen} S_{gen} + N_{WT} S_{WT} \quad (22)$$

$$C_F = \sum N_{gen} C_F \quad (23)$$

The real interest rate is computed by Fisher's formula

$$I_r = \frac{i - i_f}{(1 + i_f)} \quad (24)$$

S was estimated by applying the next formula:

$$S = C_R \left(\frac{N_{comp} - \left(N_{proj} - N_{comp} \times \text{INT} \left(\frac{N_{proj}}{N_{comp}} \right) \right)}{N_{comp}} \right) \quad (25)$$

CF is determined by applying the next equation:

$$C_F = C_{bio} \times M_{total} \quad (26)$$

The cost of generating energy (COE) for a particular system is an economic assessment of the cost of the energy generating system, incorporating associated costs in its lifetime.

$$COE = \left(\frac{C_{NPC}}{E_{served}} \right) \times \left(\frac{i \left(1 + \left(\frac{i + I_f}{1 + I_f} \right)^{R_{proj}} \right)}{\left(1 + \left(\frac{i + I_f}{1 + I_f} \right)^{R_{proj}} \right) - 1} \right) \quad (27)$$

3.2. Introduced system power management strategy

The presented methodology seeks the optimal grouping of PV modules, biomass generators, wind turbines, and batteries are mitigating the estimate of the specific EEF and LPSP. The produced power of the PV panels and wind turbines supply load demand and the battery banks are charged by the excess load, while the biomass generator is used as a backup to cover the power deficit. Fig. 2 below shows a simple flowchart of the power arrangement scenario of the introduced system.

4. Optimization methods

The paper shows the utilization of three optimization techniques to

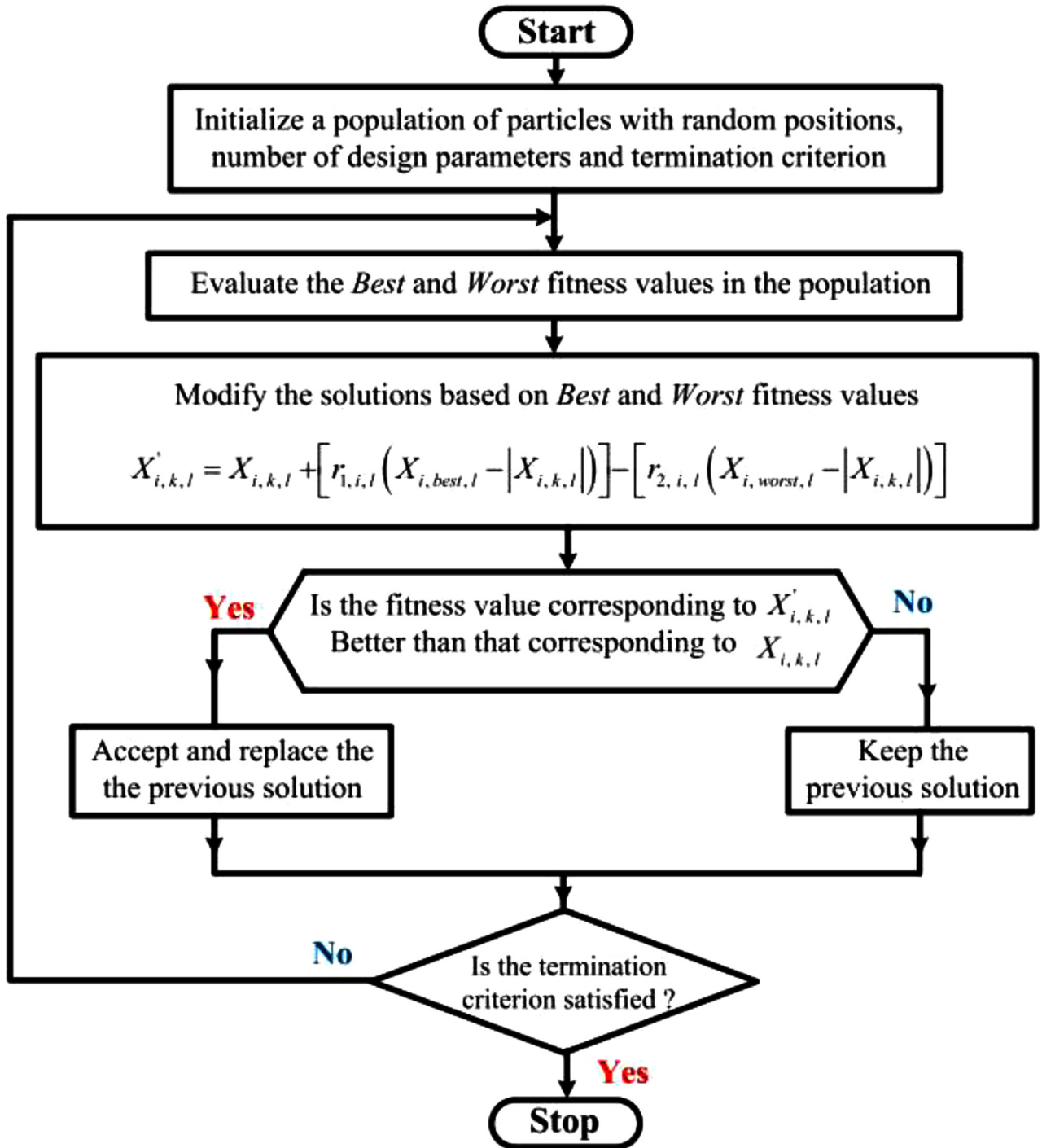


Fig. 5. The flowchart of Jaya algorithm.

resolve the problem of HRES sizing. Optimization techniques with the best efficiency and minimum operation time were directly chosen.

4.1. Particle swarm optimization (PSO)

In 1995, Kennedy and Eberhart created the PSO mechanism that was based on the natural swarm behavior (Kennedy, 2011). However, the first PSO technique used started with an arbitrarily initiated population that upgrades its places via a velocity vector. The velocity vector incorporated best suited global and personal positions. The difference between the two positions was that the global better resolution indicates the better fitness value and was discovered by gathering

particles in the swarm. The personal better resolution, on the other hand, denoted the best fitness value demonstrated by an individual particle. Fig. 3 indicates the pseudo-code of the PSO algorithm.

4.2. Harmony search (HS)

The harmony search is an optimization technique that is based on music (harmony) and was first created by ZW Geem (Geem, Kim, & Loganathan, 2001). The motivation behind the algorithm was when Geem observed that music always aims to obtain the idealistic level of harmony (Geem et al., 2001). In effect, finding harmony in music is identical to finding optimum resolution in the optimization process.

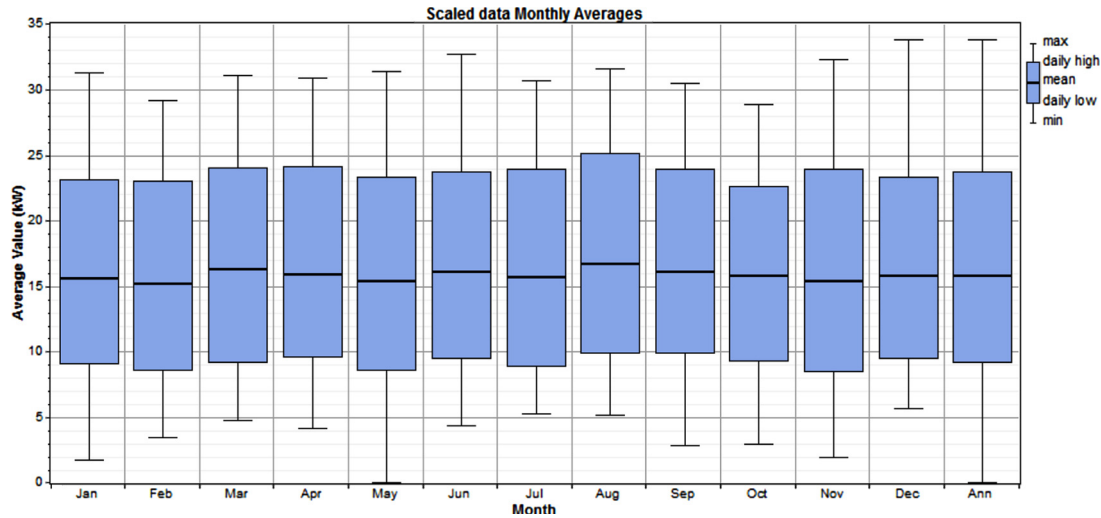


Fig. 6. The load profile of the studied zone.

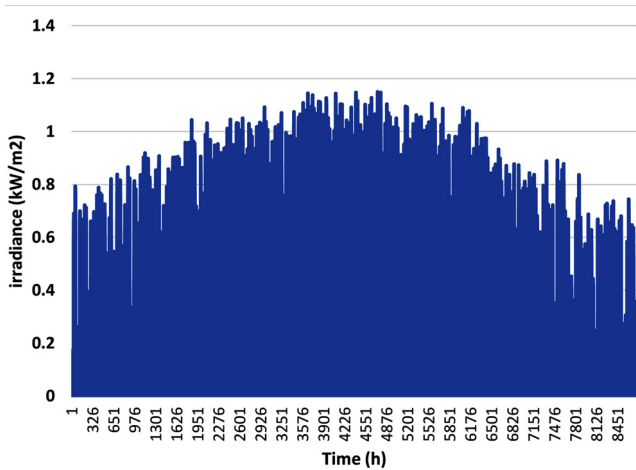


Fig. 7. Hourly insolation profile data in a year.

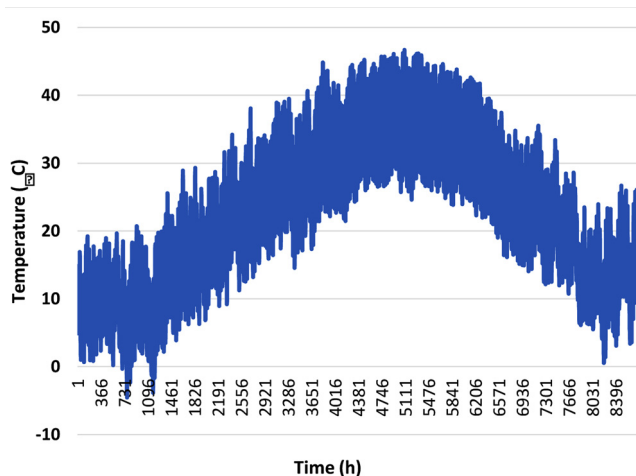


Fig. 8. Hourly ambient temperature profile data in a year.

The pseudo-code of the HS technique is introduced in Fig. 4.

4.3. Jaya algorithm

Jaya technique was developed in 2016 by Venkata Rao. It is a

heuristic algorithm that relies on swarm intelligence and applied in finding solutions for constrained and unconstrained optimization problems (Jaya, 2016). The algorithm's major idea is hinged on the belief that searching agents always look for the best solution. The best solution, in this case, is arrived at by evading the worst result at every iteration. In employing this technique, the searching agent's position updating process depends squarely on considering the best alternative only and eliminating all other positions. Thus, the solution on the preceding iteration is better than the previous one and updating the Jaya algorithm until the best solution is achieved. Fig. 5 outlines the flowchart of the Jaya algorithm.

5. Results and discussions

The present study was conducted for a site located at latitude 29° 49.794' N and longitude 39° 34.362' E. An estimation of the load profile is based on real load data. Fig. 6 shows the seasonal load profile of the area under study with a peak of 28 kW; the average load of studied area is 267 kWh/day with a load factor of 0.392.

The simulation consequences were determined by applying the MATLAB R2016a software package. Fig. 7 illustrates the dataset containing solar insolation per hour, while Fig. 8 shows the ambient temperature. For the quantity of biomass feedstock available in the village it has been assumed to be fixed and estimated at one ton per day at the price of \$ 20/ton.

Regarding the available products, the best selections in terms of operation and maintenance costs, lifetime base cost, and additional expenses have been determined. The chosen components and their corresponding costs are presented in Table 2.

To investigate the performance of the suggested hybrid system, the parameter estimation dilemma is solved by three optimization algorithms (PSO, HS, and Jaya). The results of these techniques have been compared to determine the most optimum size-based, techno-economic characteristics. The setting of the maximum and minimum bounds for decision variables is highly problem dependent (search space complexity and the number of variables). In all optimization algorithms, these bounds have been set by trial and error.

An assumption is made that the generator operates at a lower load ratio of 0.30 of its evaluated capability, which stops it from working at very low loads.

The variables of the introduced optimization techniques are:

- **PSO:** population size (n) = 25, inertia weight (w) = 0.95, inertia weight damping ratio (w_{damp}) = 0.9, personal learning coefficient (c_1) = 2.01, global learning coefficient (c_2) = 1.92, overall number

Table 2
Component characteristics and their corresponding costs.

Component	C_i (\$)	C_R (\$)	$C_{O\&M}$ (\$)	Lifetime (year)
PV (5.46 kW) which constitutes of twenty-one solar panels and two 4.4 kW fine sinusoidal wave converters.	14,854	0	10	25
400 Ah Nickel Iron Battery 48 V (40 cells @ 1.2 V each)	17,576	0	5	25
Biomass Generator (40 kW)	23,700	15,000	0.05/hour	15,000 (hours)
Wind Turbine (20 kW)	20,000	13,000	100	20

Table 3
Summarized results received from PSO, HS and Jaya.

	PV/Bio/WT/Battery			PV/Bio/Battery			PV/WT/Battery			WT/Bio/Battery		
	HS	PSO	Jaya	HS	PSO	Jaya	HS	PSO	Jaya	HS	PSO	Jaya
LPSP	1.94 %	1.92 %	1.71 %	1.92 %	1.74 %	1.60 %	2.00 %	1.32 %	1.32 %	1.83 %	1.83 %	1.97 %
F _{dump}	11.94 %	12.20 %	14.51 %	12.11 %	14.35 %	13.70 %	14.97 %	10.80 %	10.80 %	13.89 %	13.89 %	13.18 %
COE (\$/kW)	0.254	0.259	0.258	0.265	0.276	0.267	0.345	0.376	0.376	3.161	3.161	3.192
C_{NPC} (\$)	581,218	591,402	590,137	606,538	633,197	612,602	789,108	864,643	864,643	7,234,714	7,234,714	7,293,833
N_{pv}	13	10	14	16	17	16	15	17	17	0	0	0
N_{gen}	4	6	3	2	1	3	0	0	0	20	20	21
N_{WT}	1	2	1	0	0	0	2	1	1	26	26	26
N_{bat}	15	14	16	18	20	17	29	33	33	345	345	347

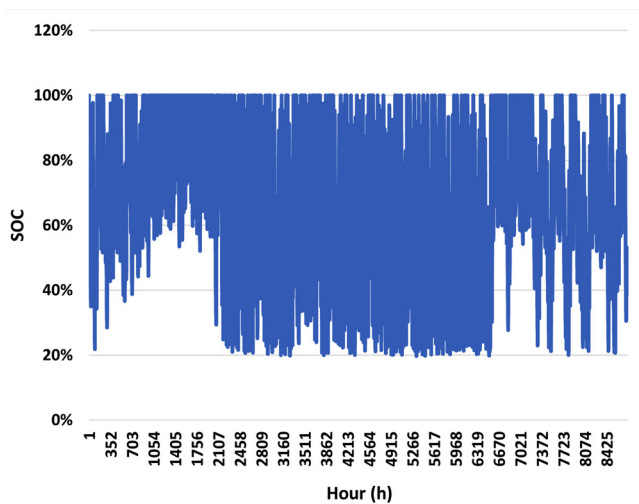


Fig. 9. Change of the battery bank SOC of the optimized PV/Biomass/Wind/battery system through the year.

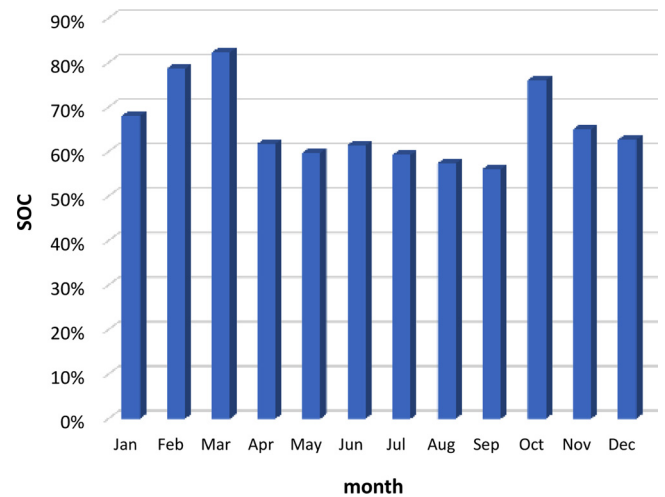


Fig. 10. The monthly average SOC of the optimized PV/Biomass/Wind/battery system.

of iterations = 100.

- **HS:** Harmony Memory Rate of 25, Number of New Harmonies of 20, Harmony Memory Consideration Rate of 0.9, Pitch Adjustment Rate of 0.1, total number of iterations = 100.

5.1. Computational results

The simulation is done for a maximum LPSP of 2 %, and a minimum EEf of 15 %. In this study, four combinations of HRES have been studied: the PV/Biomass/Wind/battery, PV/Biomass/battery, PV/Wind battery, and Biomass/Wind battery hybrid systems. This problem involves four decision parameters: N_{pv} , N_{gen} , N_{WT} , and N_{bat} .

The results of the optimal solution obtained from the three optimization algorithms for the stand-alone hybrid system are illustrated in Table 3 below.

It can be noticed from Table 3 that the fittest solution received for the HRES is the PV/Biomass/Wind/battery hybrid system that is obtained by the HS algorithm with a total cost of \$581,218 and a COE of 0.254 \$/kW. The system consists of 13 PV (70.98 kW), four biomass

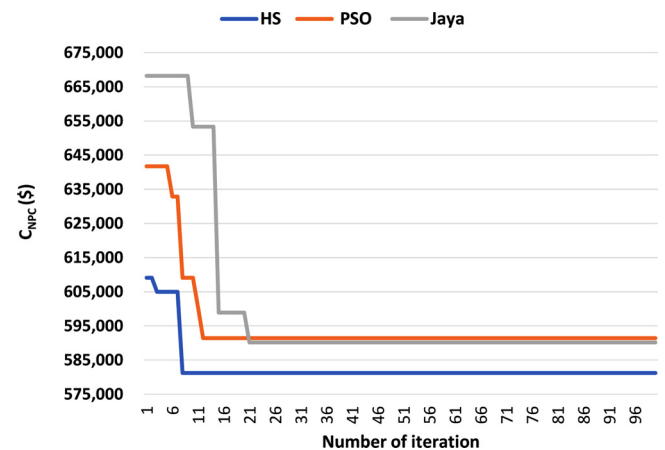


Fig. 11. The decrease of the system's net present cost the iterations for PV/Biomass/Wind/battery system.

Table 4
C_{NPV} statistics of the results from the HS, Jaya and PSO algorithms.

	HS	PSO	Jaya
Min. (\$)	581,218.00	591,402.00	590,137.00
Max. (\$)	601,580.60	669,138.35	641,727.42
Average (\$)	583,254.26	602,790.55	606,072.56
STD.	3,023.16	11,340.38	7,629.23

Table 5
Execution time statistics of the results from the HS, Jaya and PSO algorithms.

	HS	PSO	Jaya
Min. (sec.)	104.37	100.67	295.48
Max. (sec.)	125.22	124.95	309.23
Average (sec.)	109.23	113.45	303.24
STD.	4.43	10.04	4.25

Table 6
The annual GHG emissions of the optimal system.

GHG emission type (tons)	The base case	PV	Biomass power system	Wind Turbine	Net Saving
CO ₂	77.53	5.47	1.15	0.16	70.75
CH ₄	1,783.28	125.86	26.36	3.73	1,627.34
N ₂ O	24,035.58	1,696.37	355.22	50.22	21,933.77
SF ₆	1,853.07	130.78	27.39	3.87	1,691.02
Total GHG	27,749.46	1,958.49	410.11	57.98	25,322.89

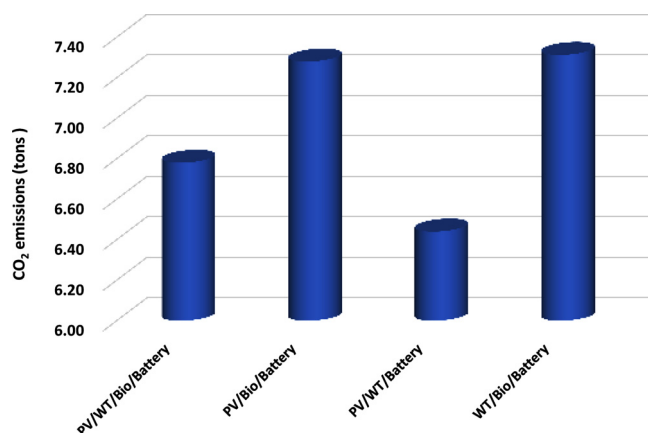


Fig. 12. The total CO₂ emissions of the four hybrid systems.

systems (160 kW), one wind turbine (20 kW) and 15 NI-Fe battery banks (288 kWh).

The predicted cost of energy is comparable to that reported in previous studies. The energy cost was reported as \$0.279/kWh for a stand-alone Wind/PV/diesel system electrifying a large resort center in the South China Sea, Malaysia (Hossain, Mekhilef, & Olatomiwa, 2017). Also, the cost of energy was reported as \$0.307/kWh for a PV-Fuel cell grid-connected system feeding a load of Electronics Faculty of the Istanbul Technical University at the Ayazaga Campus (Türkay & Telli, 2011). Similarly, for a farm in the Toshka area of Egypt a PV/Wind hybrid was designed to produce energy at a cost of \$0.597/kWh (Abdel-Salam et al., 2011). Naqvi, Yan, Dahlquist, and Naqvi (2017) have studied the viability of waste gasification based on off-grid electricity generation where the authors found that the levelized cost of electricity varies between \$0.29/kWh and \$0.40/kWh depending on the plant loading and the capacity factor.

It is assumed that the battery bank is fully charged at the beginning of simulation process, with an 80 % depth of discharge (DOD). The

battery bank state of charge (SOC) variation of PV/Biomass/Wind/battery hybrid system throughout the year is shown in Fig. 9. While the monthly average SOC is shown in Fig. 10, the rate of discharge of batteries is higher in the summer as compared to the winter.

Working time is considered one of the important deducting criterions for selecting any optimization mechanism. Also, computer efficiency plays a substantial role in executing time. Performance schematic diagrams of PSO, HS, and Jaya are demonstrated in Fig. 11. As shown in Fig. 11, HS algorithm has reached optimum value earlier than PSO and Jaya algorithms. The HS algorithm needed only nine attempts to attain optimal solution, while the PSO and Jaya algorithms took 12 and 23 attempts, respectively.

To verify the effectiveness of the HS, Jaya and PSO algorithms, a total number of 20 independent runs were performed for each algorithm. Tables 4 and 5 show the mean, maximum, minimum and standard deviation value of C_{NPV} and the execution time for the 20 runs. It is evident from the tables that the HS algorithm shows minimum deviation, which makes HS better than the Jaya and PSO algorithms.

5.1.1. Emissions analysis

The GHG emissions of the optimum hybrid systems are shown in Table 6. This table displays the amount of CO₂, CH₄, N₂O, and SF₆ emissions in tons, in addition to the total net saving of carbon dioxide by operating the system. As can be noticed from the table, the total GHG emissions from the optimal system are estimated in the simulation to be 2,427 tons. A comparison with the base case (grid-connected system) demonstrates that the utilization stand-alone PV/Biomass/Wind/battery hybrid system may decrease GHG emissions by 91.3 %.

Amongst the greenhouse gases, CO₂ has the highest portion of fuel combustion and is the chief reason for the greenhouse effect. As shown in Fig. 12, the total CO₂ emissions of the PV/WT/Bio/Battery, PV/Bio/Battery, PV/WT/Battery, and WT/Bio/Battery hybrid system are 6.78, 7.28, 6.44, and 7.31 tons, respectively. The figure also indicates that the PV/WT/Battery hybrid system has the lowest CO₂ emission, while the WT/Bio/Battery hybrid system produced the highest CO₂ emission.

6. Conclusions

This paper studied the dilemma related to the optimal sizing of hybrid renewable power systems to realize their full potential at a lower cost and also reduce the utilization of fossil fuels that produce environmental pollutants. Four hybrid systems were designed, consisting mainly of solar photovoltaics, wind turbine, biomass system, and battery bank, to determine the optimum system to fulfill the electrical requirements of a small rural community. That model was designed according to the four decision variables correlated to the system of renewable energy elements: number of PV modules, biomass generators, battery bank size, and wind turbines. The simulation was done for a maximum LPSP of 2 %, and a minimum EEF of 15 %.

The performance of two optimization algorithms, that is, HS and Jaya algorithm was judged and compared with the PSO to identify the techno-economic optimal design of the proposed hybrid system. The comparison result has shown that the harmony search algorithm presents faster convergence to the optimal solutions and is thus more efficient when determining the best solution.

The Study demonstrated that the ideal system with the least cost and the best performance was the PV/WT/Bio/Battery stand-alone hybrid system with a total system present cost of \$581,218 and a 0.254 \$/kWh cost of energy. Greenhouse gas emissions from the proposed systems were also studied. The study showed that the PV/WT/Battery hybrid system has the lowest CO₂ emission while the WT/Bio/Battery hybrid system produced the highest CO₂ emission.

Declaration of Competing Interest

The authors declare no conflict of interest with any one or organizations.

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